# Sean Luka Girgis

Senior C/C++ / Pro*C / SQL Engineer | Oracle, Unix/Linux & Legacy Modernization

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LinkedIn | GitHub | Portfolio

## Professional Summary

Senior enterprise engineer with 20+ years of experience across C, C++, Pro*C, embedded SQL, Oracle/RDBMS, Unix/Linux, multithreading, memory management, performance tuning, debugging, and legacy system modernization. Strong background developing and supporting mission-critical billing, transaction-processing, migration, and database-integrated applications using C/C++, Pro*C, PL/SQL, SQL scripts, Oracle, MySQL, DB2, Unix/Linux, and enterprise version control. Best aligned to legacy modernization and high-performance enterprise application roles requiring C/C++ systems work, database integration, SQL optimization, Unix/Linux troubleshooting, code review, technical leadership, and production support.

## Professional Experience

**Senior Capacity & Data Engineer at CITI** (2017-11 – 2025-12)

- Supported mission-critical banking infrastructure reporting and production reliability workflows using Oracle, SQL, Python, BMC TrueSight/TSCO, CA Wily/APM, AppDynamics, and AWS-backed reporting layers.
- Optimized SQL queries, Oracle-backed reporting structures, schemas, indexes, views, and data models to improve performance, maintainability, and reliability of operational reporting outputs.
- Troubleshot production issues by tracing source feeds, SQL behavior, transformation logic, missing or late data, failed outputs, latency symptoms, and downstream reporting discrepancies.
- Built validation and reconciliation checks to compare current outputs against trusted baselines, identify abnormal data movement, and prevent inaccurate reporting from reaching stakeholders.
- Created runbooks, metric definitions, data flow notes, support procedures, troubleshooting notes, and operational documentation to improve repeatability and handoff.
- Partnered with infrastructure, application, analytics, engineering, and business teams to prioritize issues, explain findings, coordinate fixes, and improve operational reliability.

**Performance Engineer at G6 Hospitality LLC** (2017-03 – 2017-11)

- Provided application performance and production visibility support for Brand.com and critical hospitality systems using Dynatrace AppMon, Gomez Synthetic Monitoring, and HP Performance Center integration.
- Investigated performance and reliability issues by correlating monitoring data, transaction behavior, and load-test results.
- Documented findings, risks, and recommendations for application and infrastructure teams supporting production-facing systems.

**SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF** (2011 – 2016)

- Supported a large financial-services application monitoring environment with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.

- Provided production troubleshooting guidance to application, middleware, infrastructure, and operations teams using CA APM/Introscope telemetry, dashboards, alerts, and transaction visibility.
- Built Perl and KornShell automation for extracting, validating, transforming, and distributing performance data, replacing manual support and reporting workflows with repeatable processes.
- Created technical documentation, support procedures, onboarding material, dashboard standards, alert thresholds, and troubleshooting notes used by multiple teams.

**Performance Test Engineer at AT&T** (2010-08 – 2011-07)

- Analyzed J2EE telecom applications under load to identify throughput limits, JDBC bottlenecks, thread contention, heap pressure, CPU constraints, memory issues, and garbage-collection symptoms.
- Configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to improve runtime visibility during testing and production-readiness reviews.
- Documented performance baselines, defects, risks, and remediation recommendations for engineering and release teams.

**Senior Systems & Data Migration Engineer at Sabre** (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while preserving transaction performance and data integrity.
- Built C++/OCCI/OCI validation and regression testing utilities to verify correctness across millions of migrated transaction records.
- Optimized C++ transaction processing and Oracle access patterns for high-volume workloads, reducing physical hardware footprint by 95% while maintaining low-latency query behavior.
- Supported schema conversion, data validation, phased cutover, troubleshooting, documentation, and production-readiness activities for a mission-critical enterprise migration.

**Architect / Developer (IRS CADE Project) at Computer Science Corporation (CSC)** (2007-10 – 2008-05)

- Designed UML class and sequence diagrams and developed modules for IRS CADE mainframe modernization.
- Built CICS, MQ Series, XML messaging interfaces, VC++ components, and DB2-backed integration modules in a government modernization environment.
- Supported technical design documentation, integration patterns, test-readiness discussions, and structured data exchange across modernization components.

**Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint** (2001 – 2007)

- Developed high-availability multithreaded C++ billing interfaces using POSIX threads, mutex locks, semaphores, socket programming, Marconi APIs, Oracle, Pro*C, PL/SQL, SQL scripts, XML, and Unix/Linux environments.
- Built and maintained AMDOCS billing, EDI, Flexible Bill Formatter, Enabler, CSM, EMS, and PRMS interfaces across invoicing, settlement, reporting, and customer-management workflows.
- Delivered performance improvements in C/C++/Pro*C/PL/SQL billing processes, reducing memory usage by 75%, improving throughput by 20%, and improving database performance by 10x through SQL, sequence, and process optimization.
- Reverse-engineered and documented undocumented AMDOCS PRM APIs to enable interface integration not directly supported by the vendor.
- Designed reusable threading and socket utility libraries using object-oriented design patterns, UML/Rational Rose, functors, containers, and C++ systems programming techniques.

- Automated WebLogic/WebSphere administration and production housekeeping using KornShell, Perl, Awk, Syncsort, SQL scripts, and Unix tools.
- Provided production support, defect investigation, data maintenance, issue escalation, troubleshooting, deployment support, and QA-cycle coordination for telecom billing and customer-management applications.
- Maintained code in CVS and ClearCase across HPUX, SunOS/Solaris, Linux, Oracle, WebLogic/WebSphere, Tuxedo/Jolt, MQ Series, and enterprise messaging environments.

## Education

**Post-Graduate Diploma in Computer Engineering Technology / Computer Science**
Humber College, Toronto, Canada

**Bachelor of Science in Civil Engineering**
Zagazig University, Egypt

## Skills

## Flagship Projects

**High-Throughput Oracle RAC Migration**
Led validation and production-readiness support for migration from 200+ MySQL nodes to a 6-node Oracle RAC platform, using C++/OCCI/OCI validation utilities and SQL optimization to preserve transaction correctness and performance.
**Technologies:** C++, OCCI, OCI, Oracle RAC, MySQL, SQL, Migration Validation

**Telecom Billing Interface Modernization**
Developed and supported multithreaded C++/Pro*C billing interfaces with Oracle, PL/SQL, XML, socket programming, Unix scripting, and production support for high-volume telecom billing systems.
**Technologies:** C++, Pro*C, Oracle, PL/SQL, POSIX Threads, Unix, XML

**AMDOCS API Reverse Engineering and Support**
Reverse-engineered undocumented AMDOCS PRM APIs, documented integration behavior, and built support utilities for billing, settlement, customer-management, and reporting workflows.
**Technologies:** C, C++, Oracle, SQL, AMDOCS, API Discovery, Documentation